Distance function 距离函数

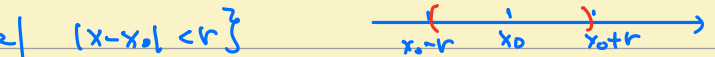
1. $|x-y| \geq 0$, $|x-y| = 0 \iff x = y$
2. $|x-y| = |y-x|$
3. $|x-y| \leq |x-z| + |z-y|$

E.g.: $d(x,y) = 2|x-y|$ ✓

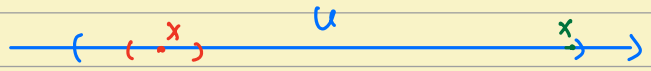
1.5 Open, closed and bounded set (开集)

Defn: Given $x_0 \in \mathbb{R}$, an **open interval centered at x_0** is a set of $\mathbb{R}$ of the form (x_0-r, x_0+r) for some $r>0$

$$(x_0-r, x_0+r) = \{x \in \mathbb{R} \mid |x-x_0| < r\}$$
$$= \{x \in \mathbb{R} \mid d(x, x_0) < r\}$$



Defn: (open set) A set $U \subseteq \mathbb{R}$ is an **open set** if for any $x \in U$, there exists $\varepsilon > 0$, such that $(x-\varepsilon, x+\varepsilon) \subset U$.



There is no $x \in U$ such that for all $\varepsilon > 0$, $\bar{x} \in (x-\varepsilon, x+\varepsilon)$ and $\bar{x} \notin U$

Defn: (closed set) A set $F \subseteq \mathbb{R}$ is a **closed set** if the complement set F^c is an open set.

闭集

A set $F \subseteq \mathbb{R}$ is a closed set if it contains all of its boundary points.

E.g.: $\mathbb{R}$ is an open set? Yes

$\emptyset$ is an open set? Yes

$\mathbb{R}$ and $\emptyset$ are Both open and closed sets.

Defn: (Bounded set) A set $S \subseteq \mathbb{R}$ is a bounded set if there exists $K \in \mathbb{R}$, such that

$$\forall x \in S, \quad |x| \le K$$

Then: ① The union of an arbitrary (任意的) collection of open sets is an open set.

② The intersection of a finite number of open sets is an open set.

infinite counterexample: $A_i = (-\frac{1}{i}, 2)$ $i \to +\infty$ $\bigcap A_i = [0, 2)$

Proof: ① Let $\{A_i\}_{i \in I}$ be a collection of open sets. Give $x \in \bigcup_{i \in I} A_i$, then there exists $i_0 \in I$ such that $x \in A_{i_0}$. Since A_{i_0} is open, there exists $\varepsilon > 0$, such that $(x_0 - \varepsilon, x_0 + \varepsilon) \subseteq A_{i_0} \in \bigcup_{i \in I} A_i$. Hence $\bigcup_{i \in I} A_i$ is open.

② $x \in \bigcap_{i \in I} A_i$ then $x \in A_i$ for all $i \in I$.

$i = 1$ $\exists\, \varepsilon_1 > 0$, $(x - \varepsilon_1, x + \varepsilon_1) \subseteq A_1$

$i = 2$ $\exists\, \varepsilon_2 > 0$, $(x - \varepsilon_2, x + \varepsilon_2) \subseteq A_2$

$\vdots$

$i = N$ $\exists\, \varepsilon_N > 0$, $(x - \varepsilon_N, x + \varepsilon_N) \subseteq A_N$

$\varepsilon := \min_{i \in I} \varepsilon_i > 0$

$\Rightarrow \exists\, \varepsilon > 0$, $(x - \varepsilon, x + \varepsilon) \subseteq A_i \quad \forall\, i \in I$

$\subseteq \bigcap_{i \in I} A_i$

Hence $\bigcap_{i \in I} A_i$ is an open set.

Thm: ① The intersection of an arbitrary collection of closed sets is a closed set.

② The union of a finite number of closed sets is a closed set.

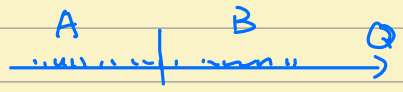
counterexample: $A_i = [0, 1-\frac{1}{i}]$ $i \to +\infty$ $\bigcup_{i \in I} A_i = [0, 1)$

实数域与有理数域的区别：完备性 complete continuous

E.g. $0.999\cdots = 1$? 极限的概念

证明：$0.999\cdots = 1$.

Dedekind's cut (戴德金分割)



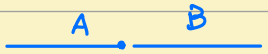
① $A \cap B = \phi$

② $A \cup B = \mathbb{Q}$

③ $\forall a \in A$, $b \in B$, we have $a < b$

There exist three different cases:

Case 1: $\sup A \in A$, and $\inf B \notin B$



E.g. $A = \{x \in \mathbb{Q}, x \le 2\}$ $B = \{x \in \mathbb{Q}, x > 2\}$

Case 2: $\sup A \notin A$, and $\inf B \in B$

E.g. $A = \{x \in \mathbb{Q}, x < 2\}$ $B = \{x \in \mathbb{Q}, x \ge 2\}$

Case 3: $\sup A \notin A$, and $\inf B \notin B$

E.g. $A = \{x \in \mathbb{Q}, x < 0$, or $x^2 < 2\}$

$B = \{x \in \mathbb{Q}, x > 0$, and $x^2 > 2\}$

Case 4: $\sup A \in A$, and $\inf B \in A$ → impossible

Consider a cut : case1 $A \mid B \to 0.999\cdots$

case2 $C \mid D \to 1$

$A = \{x \in \mathbb{Q},\ x < 0.999\cdots\}$

$C = \{x \in \mathbb{Q},\ x < 1\}$

We need to prove: $A = C \Leftrightarrow A \subseteq C$ and $C \subseteq A$

Proof: ① $A \subseteq C$ take $x \in A$, $x < 0.999\cdots \Rightarrow x < 1, \Rightarrow x \in C$

② $C \subseteq A$. take $x \in C$, then $x < 1$, since $x \in \mathbb{Q}$, we have $\exists p, q \in \mathbb{Z}$, $x = \frac{p}{q} < 1$. $\Rightarrow p < q$.

$1 - x = 1 - \frac{p}{q} = \frac{q-p}{q} \geq \frac{1}{q}$ because $p, q \in \mathbb{Z}$

there exists $n > N$ such that $10^n > q \Rightarrow \frac{1}{10^n} < \frac{1}{q}$

$1 - x = \frac{q-p}{q} \geqslant \frac{1}{q} > \frac{1}{10^n} \Rightarrow 1 - x > \frac{1}{10^n}$

$\Rightarrow x < 1 - \frac{1}{10^n} = \underbrace{0.999\cdots}_{\text{有 } n \text{ 个 } 9} < \underbrace{0.99\cdots}_{+\infty}$

Hence $x \in A$ Therefore $A = C \Rightarrow 0.99\cdots = 1$.

Limits of functions

Notation: the limit of fumtion f at x_0

$\lim\limits_{x \to x_0} f(x)$ $\lim\limits_{n \to +\infty} f(x_n)$ $n \in \mathbb{N}_+$

the limit of function f as x_0 goes to infinity

$\lim\limits_{x \to +\infty} f(x)$ $\lim\limits_{x \to -\infty} f(x)$

Defn (ε-δ-definition) Let $f: \mathbb{R} \to \mathbb{R}$ is a real-valued function. Suppose that $x_0 \in (a,b)$ and f is well-defined on (a,b) except possibly (可能不是) at x_0, then the limit of f at x_0 is A, if and only if:

For any $\varepsilon > 0$, there exists $\delta > 0$, such that

$$|f(x) - A| < \varepsilon \quad \forall \ 0 < |x - x_0| < \delta$$

$f(x) \to A$

$x \to x_0$

E.g.: The derivative (导数)

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

$f(x) = x^2$

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

$$= \lim_{h \to 0} \frac{(x_0+h)^2 - x_0^2}{h} = \lim_{h \to 0} \frac{x_0^2 + 2x_0 h + h^2 - x_0^2}{h}$$

$$= \lim_{h \to 0} \frac{2x_0 \cdot h + h^2}{h} \qquad \lim_{h \to 0} 2x_0 + h = 2x_0$$

E.g.: show that by definition:

$$\lim_{x \to 8} f(x) = x = 8$$

任取给的 ε，都存在找到 δ，使函数逼近的值与 x 趋近的值之间误差的关系.

Proof: For any $\varepsilon > 0$, there exists $\delta > 0$, such that if $|x - 8| < \delta$, we have $|f(x) - 8| = |x - 8| < \varepsilon$, then we can take $\delta = \varepsilon$.

$$\lim_{x\to 8} f(x) = 10x-2 = 78$$

Proof: For any $\varepsilon>0$, there exists $\delta>0$, such that if $|x-8|<\delta$, $\Rightarrow$ $|10x-2-78| = 10|x-8| < \varepsilon \Rightarrow |x-8| < \frac{\varepsilon}{10}$

Then we can take $\delta = \frac{\varepsilon}{10}$, then by definition $\lim_{x\to 8} f(x) = 78$.

E.g.: $f(x) = \frac{x^2-4}{x-2}$. show that $\lim_{x\to 2} f(x) = 4$

$x \in (2-\varepsilon, 2+\varepsilon)$

Proof: For any $\varepsilon>0$, $|f(x)-4| = \left|\frac{x^2-4}{x-2} - 4\right|$

suppose $x \neq 2$, then

$\left|\frac{x^2-4}{x-2} - 4\right| = |x+2-4| = |x-2| < \varepsilon$, then there exists $\delta = \varepsilon$, such that if $|x-2|<\delta$, $|f(x)-4|<\varepsilon$, by definitions

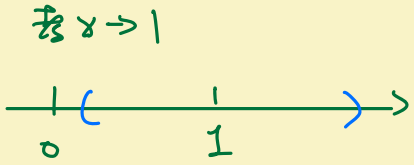
$$\lim_{x\to 2} f(x) = 4$$

show that $\lim_{x\to 1} \frac{x^2-1}{2x^2-x-1} = \frac{2}{3}$

Proof: For any $\varepsilon>0$, $\left|\frac{x^2-1}{2x^2-x-1} - \frac{2}{3}\right| = \left|\frac{(x+1)(x-1)}{(2x+1)(x-1)} - \frac{2}{3}\right|$

Suppose $x \neq 1$, $\Rightarrow \left|\frac{x+1}{2x+1} - \frac{2}{3}\right| = \left|\frac{3(x+1)-2(2x+1)}{3(2x+1)}\right|$

$$= \frac{|x-1|}{3|2x+1|} < \varepsilon$$



Let $|x-1|<1 \Rightarrow x>0 \Rightarrow |2x+1|>1$

Then we can take $\delta = \min\{3\varepsilon, 1\}$, because

$$\frac{|x-1|}{3|2x+1|} < \frac{|x-1|}{3} < \varepsilon, \text{ if } |x-1|<\delta$$

then we have $\lim\limits_{x\to 1} f(x) = \frac{2}{3}$

Note:

① δ 依赖于 ε，ε 越小，则 δ 越小，但 δ 取更小也无妨.

② 只要求在 $(x_0-\delta, x_0)\cup(x_0, x_0+\delta)$ 内，$f(x)$ 有定义即可，不要求在 $x=x_0$ 处有定义

Thm (Squeeze theorem, or Sandwich theorem, 夹逼定理)

Suppose $\delta>0$, for all $x\in(x_0-\delta, x_0)\cup(x_0, x_0+\delta)$, such that $f(x)\le h(x)\le g(x)$.

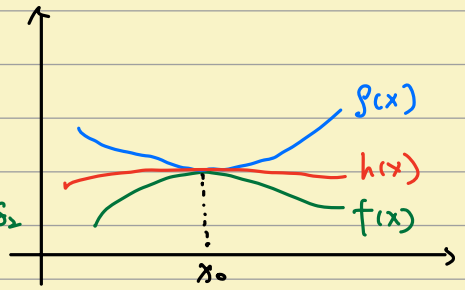
If $\lim\limits_{x\to x_0} f(x) = \lim\limits_{x\to x_0} g(x) = A$, then $\lim\limits_{x\to x_0} h(x) = A$.

Proof: For any $\varepsilon>0$, there exists $\delta_1>0$, such that $|x-x_0|<\delta_1 \Rightarrow |f(x)-A|<\varepsilon$.

$\Rightarrow A-\varepsilon < f(x) < A+\varepsilon$

By similar argument, $\exists\, \delta_2>0$, $|x-x_0|<\delta_2$

$\Rightarrow A-\varepsilon < g(x) < A+\varepsilon$



Take $\delta = \min(\delta_1, \delta_2)$, we have

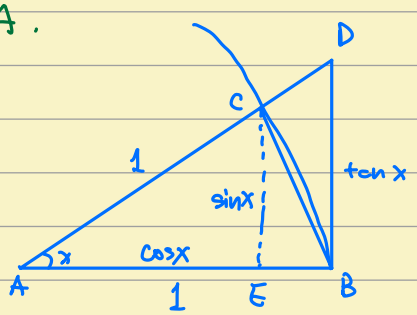
$$|x-x_0|<\delta \Rightarrow \begin{cases} A-\varepsilon < f(x) < A+\varepsilon \\ A-\varepsilon < g(x) < A+\varepsilon \end{cases}$$

Since we have $f(x)\le h(x)\le g(x)$ $\forall\, x\in(x_0-\delta, x_0)\cup(x_0, x_0+\delta)$

$\Rightarrow A-\varepsilon \le f(x) \le h(x) \le g(x) \le A+\varepsilon$

$\Rightarrow A-\varepsilon \le h(x) \le A+\varepsilon \quad \Leftrightarrow |h(x)-A| < \varepsilon$

By definition, we conclude $\lim_{x\to x_0} h(x) = A$.



E.g.: Prove $\lim_{x\to 0} \frac{\sin x}{x} = 1$

Proof: $\triangle ABC = \frac{\sin x}{2}$

$\triangle ABD = \frac{\tan x}{2}$

$\overset{\frown}{ABC} = \frac{x}{2}$

$\triangle ABC < \overset{\frown}{ABC} < \triangle ABD$

$\Rightarrow \frac{\sin x}{2} < \frac{x}{2} < \frac{\tan x}{2}$

$\Rightarrow \sin x < x < \tan x = \frac{\sin x}{\cos x}$

$\Rightarrow 1 < \frac{x}{\sin x} < \frac{1}{\cos x} \Rightarrow \cos x < \frac{\sin x}{x} < 1$

since $\lim_{x\to 0} \cos x = 1 \quad \lim_{x\to 0} 1 = 1$

$\lim_{x\to 0} \frac{\sin x}{x} = 1$ by squeeze them.

Ex. show that $\lim_{x\to 8} x^2 = 64$